

Falsifiability and Experimental Frontiers of Phase Theory

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Abstract

A foundational physical theory must be empirically vulnerable. The history of physics is, in large part, a history of theories that placed themselves before nature and accepted the verdict. In this paper, we identify explicit falsification criteria and experimentally accessible frontiers for Phase Theory — a framework in which phase is the sole fundamental constituent of physical reality, and in which space, time, matter, light, and geometry are emergent from the global organization, topology, and consistency of a single phase-bearing substrate. We delineate regimes in which Phase Theory predicts behavior indistinguishable from established frameworks — quantum mechanics, general relativity, quantum field theory, and thermodynamics — as well as regimes in which it predicts distinct, testable deviations arising from phase admissibility constraints, coherence limits, and saturation effects. We specify seven experimental test classes: randomness collapse, non-collapse measurement memory, photon non-paradox, phase cooling, black hole information leakage, cosmological phase drift, and information density bounds. For each class, we enumerate the falsification condition precisely. We additionally enumerate four explicit negative predictions — what Phase Theory asserts will not be observed — and provide a risk matrix summarizing the theory's principal empirical vulnerabilities. Phase Theory is compared to many-worlds, string theory, and dynamical collapse models on the criterion of falsifiability. The paper establishes Phase

Theory as a scientifically accountable framework that accepts, rather than deflects, the burden of experimental judgment.

1. Why Falsifiability Must Be Explicit

Karl Popper's criterion of falsifiability remains, more than six decades after its canonical formulation [1], the sharpest available boundary between scientific theory and metaphysical speculation. A claim about the physical world is scientific if and only if it specifies conditions under which it would be shown to be false. This is not a sociological convention or an arbitrary demarcation device — it reflects a deep epistemological asymmetry: no finite collection of confirming instances can establish a universal law, but a single well-attested counterinstance can refute one. Physics has prospered by accepting this asymmetry honestly. Newtonian mechanics was not abandoned because it lacked elegance, but because it failed to account for the precession of Mercury's perihelion. The Michelson-Morley experiment did not merely fail to detect the luminiferous ether — it refuted its existence. The history of physics is a graveyard of theories that were not wrong because they were poorly conceived, but because nature said otherwise. This is a feature, not a defect, of the scientific enterprise.

Phase Theory makes three radical structural claims that place it in unusual conceptual territory relative to contemporary foundational frameworks. First, it eliminates fundamental randomness from the ontological furniture of physics. In Phase Theory, measurement outcomes are not governed by irreducible probabilistic law; they arise instead from the admissibility structure of phase configurations — from constraints on which global phase arrangements are internally consistent — and apparent randomness reflects an averaging over phase configurations under the coherence limitation rather than any brute indeterminacy in nature. Second, Phase Theory removes wavefunction collapse as an ontological primitive. Decoherence,

reinterpreted as an emergent phase-coherence phenomenon in which the coherence of a subsystem's phase topology is degraded by interaction with its environment, replaces collapse entirely; there is no preferred measurement basis in the fundamental description, and no discontinuous dynamical process invoked to explain the transition from superposition to outcome. Third, Phase Theory rejects both fields and spacetime as ontological primitives, deriving both from the topology and global consistency of a single phase-bearing substrate Φ : the metric tensor, connection coefficients, and curvature of spacetime emerge as collective variables describing the large-scale strain and coherence geometry of the phase substrate, while quantum fields emerge as the linearized excitation spectrum of topological defects propagating through it.

Such radical structural claims carry a correspondingly heightened burden of falsifiability. A theory that reinterprets the entire ontological apparatus of modern physics while making no prediction that cannot equally well be accommodated by existing frameworks is not physics — it is philosophy of physics, and must be evaluated on philosophical rather than empirical grounds. Phase Theory explicitly declines this safe harbor. It accepts the burden of falsifiability as a constitutive commitment rather than an external imposition, and it accepts it precisely because its structural claims are ambitious. The more radical the ontological revision proposed, the more vulnerable the theory must be willing to make itself to empirical scrutiny.

The organizing question of this paper is correspondingly simple: *How could Phase Theory be wrong?* The paper proceeds by first establishing the domains of empirical equivalence in which Phase Theory must reproduce existing results exactly (Section 2); then identifying the admissibility frontier where genuine deviations are expected (Section 3); then specifying seven test classes with explicit falsification conditions (Sections 4–10); then enumerating four negative predictions (Section 11); then providing a risk

matrix of principal vulnerabilities (Section 12); then comparing Phase Theory with competing foundational frameworks on the criterion of falsifiability (Section 13); then arguing that current experimental capabilities have opened the window for these tests (Section 14); then specifying definitive failure conditions (Section 15); and finally concluding with a call for experimental programs (Section 16).

2. Domains of Empirical Equivalence

Phase Theory does not propose to replace the predictive apparatus of quantum mechanics, general relativity, quantum field theory, and thermodynamics in the domains where those frameworks have been comprehensively validated. It proposes to subsume them — to derive their equations, symmetries, and predictions from deeper axioms, and in so doing to reveal the limits of their applicability. But derivation is a two-edged sword: a theory that derives quantum mechanics must also reproduce quantum mechanics, in all of its quantitative detail, in every regime where quantum mechanics has been experimentally confirmed. This is not aspirational — it is a hard constraint. In Phase Theory, the empirical equivalence in established domains follows as a theorem from the axioms, not as an adjustable feature [PT].

In the regime of low-energy quantum experiments — isolated microscopic systems evolving in slowly varying backgrounds, prepared and measured with standard apparatus — the Schrödinger equation emerges from the linearized phase-relaxation equation of Phase Theory in the limit of isolated topological defects propagating through backgrounds varying on scales large compared to the coherence length. The phase-relaxation equation, written schematically as

$$(1) \partial_t \Phi = -\gamma \cdot \delta I[\Phi]/\delta \Phi + \eta[\Phi]$$

reduces, under linearization around a slowly varying background Φ_0 and in the absence of topological-defect creation events, to the standard Schrödinger equation for the defect wavefunction, with the reduced Planck constant $\hbar$ emerging as a ratio of phase-substrate parameters [PT, Sec. XVII]. The Born rule for measurement probabilities is recovered as the leading-order term in an expansion in the admissibility correlation function $C_{\text{adm}}(\tau)$ — which vanishes in the linearized regime, leaving standard Born-rule statistics [PT, Sec. XVII.4].

At the level of particle physics, Standard Model cross sections and decay rates emerge from the effective dynamics of topological defects in Phase Theory's target manifold $T = U(1) \times SU(2) \times SU(3)$. The gauge symmetries of the Standard Model are not imposed as external requirements but arise as the continuous symmetry group of maps from the phase substrate into the target manifold that leave the action

$$(2) \ I[\Phi] = \int d^4x \sqrt{-g} \cdot \mathcal{L}(\Phi, \partial\beta\Phi, g_{\mu\nu})$$

invariant [PT, Sec. XIV.2]. The empirical success of the Standard Model in accounting for collider data from LEP, the Tevatron, and the LHC is therefore subsumed rather than contradicted: these experiments probe a regime in which the effective topological-defect description is excellent and deviations from it are suppressed by powers of (E/E_{coh}) , the ratio of experimental energy to the phase-coherence scale.

In the regime of weak gravitational fields, the Einstein field equations and their Newtonian weak-field limit are recovered as the equations of motion for the emergent metric

$$(3) \ g_{\mu\nu}[\Phi] = \eta_{\mu\nu} + \kappa_{\text{eff}} \cdot \sigma_{\mu\nu}[\Phi]$$

in the weak-strain limit $\sigma_{\mu\nu} \ll \eta_{\mu\nu}$, where $\sigma_{\mu\nu}[\Phi]$ is the phase strain tensor and κ_{eff} is an effective coupling derived from phase-substrate parameters [PT, Sec.

XV.3]. The precision tests of general relativity — Mercury's perihelion precession, light deflection, gravitational redshift, the Shapiro delay, and the binary pulsar timing measurements confirmed by LIGO [13] — all probe this weak-strain regime, and all are reproduced exactly.

In the domain of laboratory thermodynamics, the second law of thermodynamics and the Boltzmann entropy formula are derived from the monotonic directionality of phase relaxation: the phase-substrate functional $I[\Phi]$ decreases monotonically under relaxation dynamics (Theorem 21.1 of [PT]), and this monotonicity is identified with the increase of thermodynamic entropy in macroscopic systems [PT, Sec. XXI]. Carnot efficiencies, fluctuation theorems, and the full apparatus of equilibrium and non-equilibrium statistical mechanics are recovered as limiting cases [PT, Sec. XXI.4].

These four domains of empirical equivalence are not a menu from which Phase Theory selects favorable items — they are a hard constraint. Any measurable deviation from established quantum mechanics in a low-energy isolated-system experiment, from Standard Model cross sections at collider energies below the coherence scale, from general relativity in the weak-field regime, or from thermodynamics in any laboratory setting would immediately falsify Phase Theory. The theory is constrained to reproduce all of these results exactly, and it accepts this constraint explicitly as a condition of its own coherence.

3. Where Deviations Are Expected: The Admissibility Frontier

Phase Theory predicts deviations from standard frameworks only when admissibility constraints become dynamically active — that is, when experimental conditions approach the boundaries of what the phase-substrate axioms permit. These boundaries are not arbitrary cutoffs introduced by hand to avoid conflict with existing data; they are structural

consequences of the axioms, particularly of the coherence limitation (Axiom 4 of [PT]), which asserts that the mutual information between any two subsystems A and B of the phase substrate is bounded by the area of their shared boundary:

$$(4) J(\Phi_A; \Phi_B) \leq C \cdot \min(|\partial A|, |\partial B|)$$

where C is a phase-substrate constant with dimensions of (information per unit area) and $|\partial A|$, $|\partial B|$ denote the boundary areas of the respective subsystems [PT, Axiom 4]. This inequality is the phase-theoretic origin of holography [10, 11, 12], and its activation — when the system approaches the saturation point $J \approx C \cdot |\partial A|$ — is the primary source of observable deviations.

The admissibility frontier is crossed under precisely four experimental conditions. The first occurs when phase coherence approaches saturation: as a system is engineered to approach the maximum mutual information permitted by Axiom 4, the effective dynamics depart from standard quantum behavior in ways parameterized by the admissibility correlation function $C_{\text{adm}}(\tau)$. In the standard quantum regime, this function is identically zero; approaching saturation, it acquires a non-zero value that scales as the fractional approach to the coherence bound.

The second condition under which the admissibility frontier becomes active occurs when distinguishability limits are systematically tested. Phase Theory assigns to each measurement outcome a phase-sector label that is constrained by the admissibility structure of the phase substrate. When experimental conditions are engineered to approach the maximum distinguishability of adjacent phase sectors — that is, when the phase gradient $\nabla\Phi$ at the measurement boundary approaches the maximum permitted value — the admissibility structure leaves an empirical trace in the outcome statistics that is absent from standard quantum mechanical predictions. Specifically, outcomes in adjacent phase sectors exhibit

correlations parameterized by the sector boundary width, which is not a free parameter but is determined by the phase-substrate equation of state.

The third condition arises in the regime where non-field descriptions dominate the dynamics. Near the cores of topological defects — in the interior of the mathematical objects that Phase Theory identifies with particles — the field-theoretic description that approximates defects as point particles in a smooth background field breaks down. The full phase-topological dynamics must be used, and these dynamics predict behavior near defect cores that differs from standard quantum field theory in ways that are, in principle, accessible in experiments probing sub-coherence-length structure. This regime is the most technically challenging to access, as it requires either energies approaching the coherence scale or precision measurements at the defect-core scale.

The fourth condition under which deviations appear is when phase restructuring occurs faster than thermodynamic equilibration. Standard open-system quantum mechanics predicts that coherence is degraded by interaction with a thermal bath on a timescale set by the system-bath coupling and the bath temperature. Phase Theory predicts that an additional mechanism — admissibility-structured initialization — can preserve coherence at rates lower than those predicted by the standard thermal decoherence formula, because the phase-topological structure of the initialized state is constrained to minimize mutual information leakage per unit boundary area. When this condition is engineered experimentally, the system can exhibit what Phase Theory calls non-thermal phase memory effects: coherence preservation at rates inconsistent with the thermal baseline of standard open-system theory.

Each of these four conditions is experimentally rare but reachable with current or near-future technology. The seven test classes that follow are

organized around the specific experimental setups best suited to approach and probe each admissibility frontier regime.

4. Test Class I: Admissibility Structure in Outcome Statistics

The Born rule — the assertion that the probability of obtaining outcome k in a measurement on a quantum system in state $|\psi\rangle$ is $P(k) = |\langle k|\psi\rangle|^2$ — is perhaps the most precisely tested quantitative law in all of physics. Its experimental confirmation extends across an enormous range of systems, energies, and conditions [9]. Standard quantum mechanics asserts that this rule is exact and that the randomness it encodes is irreducible: there is no deeper structure from which Born-rule probabilities emerge as approximations to something more determinate.

Phase Theory contests this interpretation at a structural level. It asserts that Born-rule statistics are exact in the linearized regime — when the admissibility correlation function $C_{\text{adm}}(\tau)$ vanishes — but that they are approximations to a more structured distribution in the admissibility frontier regime, where $C_{\text{adm}}(\tau) \neq 0$. The physical content of this prediction is as follows. In a long sequence of measurements performed under precisely controlled coherence depth — conditions engineered to approach the saturation bound of Equation (4) — individual outcomes are not independent Bernoulli trials. They exhibit cross-correlations on a timescale τ_{coh} set by the coherence depth of the phase substrate, parameterized by:

$$(5) \ C_{\text{adm}}(\tau) = \langle \delta P_k(t) \cdot \delta P_k(t + \tau) \rangle \neq 0, \quad \text{for } |\tau| < \tau_{\text{coh}}$$

where $\delta P_k(t)$ denotes the deviation of the empirical outcome probability from its Born-rule expectation at time t . In standard quantum mechanics, $C_{\text{adm}}(\tau) = 0$ for all τ , because successive measurement outcomes are statistically independent. In Phase Theory, $C_{\text{adm}}(\tau) \neq 0$ in the admissibility frontier regime, and the timescale τ_{coh} over which it decays to zero is determined by the

coherence depth of the experimental setup — a quantity that can be varied systematically as an experimental control.

The experimental path to testing this prediction requires three technical capabilities. First, ultra-low-noise photonic interference setups with sub-shot-noise phase control, enabling the experimenter to tune the coherence depth of the measurement environment with precision. Second, photon-number-resolving detectors with sufficient efficiency to avoid any spurious correlation between detection events arising from classical noise. Third, outcome sequences of sufficient length to achieve the statistical power required to distinguish $C_{\text{adm}}(\tau) \neq 0$ from $C_{\text{adm}}(\tau) = 0$: Phase Theory estimates that sequence lengths of $N > 10^8$ events per coherence-depth setting are required to achieve a 5σ detection threshold at the predicted correlation strength, which is small in the accessible regime and scales as the fractional approach to the coherence saturation bound.

The experimental protocol requires systematic variation of coherence depth as a control, because Phase Theory predicts that $C_{\text{adm}}(\tau) \rightarrow 0$ as the system is moved away from the admissibility frontier. This variation is essential for distinguishing a genuine admissibility-structure signal from any classical noise source that might contaminate the measurement: classical noise correlations do not exhibit the coherence-depth dependence predicted by Phase Theory, whereas the admissibility correlation function does. A convincing positive detection would require $C_{\text{adm}}(\tau)$ to be non-zero at high coherence depth and to decrease systematically as coherence depth is reduced, in quantitative agreement with the Phase Theory prediction.

The falsification condition for Test Class I is correspondingly precise: if outcome statistics remain perfectly consistent with Born-rule randomness under all admissibility-controlled conditions across all accessible coherence depths — that is, if $C_{\text{adm}}(\tau) = 0$ within experimental precision for all τ and all coherence-depth settings — then Phase Theory's central claim that

randomness is apparent rather than irreducible is empirically inaccessible, and the theory fails as physics rather than as philosophy. This is a clean, unambiguous failure condition that requires no reinterpretation.

5. Test Class II: Sequential Measurement Correlations Beyond Standard QM

Standard quantum mechanics, augmented by the full apparatus of decoherence theory [9], provides a complete account of sequential measurement correlations. In the standard framework, sequential measurements on independent systems are statistically independent after accounting for any entanglement between them, and correlations between successive measurement outcomes on the same system are fully accounted for by the quantum state update rule combined with decoherence-induced einselection. The Markov property of measurement sequences — that future outcomes depend on past outcomes only through the current quantum state — is a theorem of standard quantum mechanics for any system whose interaction with its environment is adequately modeled by a Lindblad master equation.

Phase Theory makes a prediction that violates this Markov property in a specific and controlled way. It asserts that the measurement process does not merely update the quantum state of the measured system — it leaves a phase-topological imprint on the phase substrate in the vicinity of the measurement apparatus, an imprint that Phase Theory calls admissibility memory. This memory is not encoded in the standard quantum degrees of freedom and therefore cannot be accounted for by any decoherence model operating within the standard quantum Hilbert-space framework. It produces context-sensitive correlations across sequential measurements — correlations that persist even when standard QM predicts full statistical independence — with a characteristic length scale λ_{adm} set by the coherence domain of the apparatus-system coupling:

$$(6) \lambda_{adm} = \hbar_{eff} / (m_{eff} \cdot v_{coh})$$

where $\hbar_{eff}$, m_{eff} , and v_{coh} are effective phase-substrate parameters characterizing the apparatus-system coupling regime [PT, Sec. XVIII.3]. These correlations are absent in standard QM and, crucially, absent in pilot-wave (Bohmian) theory as well, because pilot-wave theory shares the standard quantum state update rule. They are distinguishable from decoherence-model effects by their systematic dependence on apparatus coupling strength: as apparatus-system coupling is varied without altering the decoherence rate (by compensating temperature changes), Phase Theory predicts that the non-Markovian admissibility-memory correlations scale with coupling strength while the decoherence-induced correlations remain constant.

The experimental path for Test Class II is built around weak-measurement chains on photonic qubits. Weak measurements — measurements that extract partial information about a quantum system without fully collapsing its state [9] — are ideally suited to probing admissibility memory because they allow the sequential measurement chain to be extended over many steps while preserving quantum coherence in the system. The Phase Theory prediction is that, in a sufficiently long weak-measurement chain, the joint distribution of outcomes exhibits non-Markovian correlations with characteristic length λ_{adm} that cannot be attributed to any standard decoherence mechanism. Delayed-choice variants of this experiment — in which the measurement basis at each step is chosen after the preceding measurement has been completed — provide additional discriminating power because they rule out any causal account of the correlations that invokes signaling from earlier to later measurements.

The falsification condition for Test Class II is the following: if all correlations across sequential measurements — across all chain lengths, apparatus coupling strengths, and coherence configurations tested — are fully

accounted for by standard quantum mechanics augmented by decoherence theory, with no residual non-Markovian admissibility signature surviving systematic variation of apparatus coupling strength, then Phase Theory is falsified on this prediction class. The key experimental requirement is that the decoherence rate and the apparatus coupling strength must be independently varied, which is technically achievable in trapped-ion and superconducting systems where coupling and dissipation can be tuned with high precision.

6. Test Class III: Phase-Global Constraint Effects in Macroscopic Interferometry

Wave-particle duality — the accommodation of both wavelike interference and particle-like detection in the same physical object — is the central paradox of quantum optics and has been experimentally explored since Taylor's earliest single-photon interference experiments in 1909. Within standard quantum mechanics, this duality is framed through Bohr's complementarity principle: no experimental arrangement can reveal both wave and particle aspects of a quantum system simultaneously. Within Phase Theory, the duality is reinterpreted without paradox: what quantum mechanics calls the wave behavior of a photon is the propagation of a coherent phase mode in the phase substrate, and what it calls the particle behavior is the behavior of a topological defect localized in the substrate. These are two aspects of the same phase-substrate configuration, not two mutually exclusive descriptions requiring a complementarity principle to prevent contradiction.

However, Phase Theory goes beyond this ontological reinterpretation to make a specific quantitative prediction. It asserts that single-photon behavior at large spatial separations — specifically, in interferometers whose baseline length L is large compared to any apparatus coherence length — is governed by global phase constraints imposed by the consistency requirements of the phase substrate. These global constraints produce effects on the fringe

visibility that cannot be attributed to local hidden variables (ruled out by Bell tests [9]) or to local field fluctuations (which decay on scales set by the coherence length). Specifically, the visibility $V(L)$ of interference fringes as a function of baseline length L is predicted to exhibit a non-monotonic signature:

$$(7) \ V(L) = V_0 \cdot [1 - \varepsilon(L/L_{\text{coh}}) + \delta_{\text{PT}}(L/L_{\text{glob}})]$$

where V_0 is the zero-baseline visibility, $\varepsilon(L/L_{\text{coh}})$ is the standard decoherence-induced visibility reduction on the coherence length L_{coh} , and $\delta_{\text{PT}}(L/L_{\text{glob}})$ is a Phase Theory correction term arising from global phase-consistency enforcement at the global coherence scale L_{glob} [PT, Sec. XVI.2]. In standard quantum optics, the third term is absent: $V(L)$ is a monotonically decreasing function of L , reduced only by environmental decoherence. Phase Theory predicts a non-monotonic deviation at $L \sim L_{\text{glob}}$ — a partial recovery of visibility arising from the global phase-consistency constraint — that is a signature of the non-local phase-substrate structure with no analog in any local description.

The experimental path for Test Class III involves long-baseline free-space or fiber-based single-photon interferometers with systematic variation of baseline length from laboratory scale (~ 1 m) through intermediate scales (~ 100 m, ~ 1 km) to the longest scales achievable with current photonic technology. At each baseline length, the fringe visibility must be measured with sufficient precision to resolve the Phase Theory deviation δ_{PT} from the standard decoherence envelope ε . Adaptive boundary manipulation — adjusting the phase relationship between the two interferometer arms during photon propagation — provides an additional degree of experimental control that discriminates between genuinely global phase constraints and any local source of the visibility recovery. Phase-sensitive detection without energy exchange with the environment (homodyne detection in a carefully shielded

fiber network) is essential to avoid introducing classical phase noise that would mask the signal.

The falsification condition for Test Class III is the following: if photon behavior at all accessible baseline lengths remains fully and quantitatively described by standard quantum optics, including all known decoherence mechanisms, with no non-monotonic visibility signature at any scale, then Phase Theory is falsified on this prediction class.

7. Test Class IV: Admissibility-Engineered Coherence Beyond Thermal Limits

One of the most precisely confirmed predictions of open-system quantum mechanics is that decoherence rates scale with temperature. A quantum system coupled to a thermal bath at temperature T loses coherence at a rate Γ that, in the Markovian limit of the Caldeira-Leggett model and its descendants, satisfies

$$(8) \Gamma_{th} = \gamma_0 \cdot coth(\hbar\omega/2k_B T)$$

where γ_0 is the zero-temperature damping rate, ω is the system frequency, k_B is Boltzmann's constant, and T is the bath temperature [9]. At sufficiently high temperatures ($\hbar\omega \ll k_B T$), this becomes $\Gamma_{th} \approx 2\gamma_0 k_B T / \hbar\omega$, linear in temperature. The practical consequence of this relationship for quantum computing is well known: maintaining coherence requires physical cooling, and no coherence management strategy short of refrigeration can overcome the thermal decoherence rate at fixed bath temperature.

Phase Theory predicts a departure from this picture in a specific and testable regime. It asserts that the decoherence rate experienced by a quantum system is not determined solely by temperature and system-bath coupling, but also by the phase-topological structure of the initial state. A suitably designed phase-topological initialization (PTI) structure — a phase

configuration specifically engineered to minimize mutual information leakage per unit boundary area, saturating the admissibility bound from below rather than from above — interacts with the thermal bath at a lower effective coupling rate than a generic initial state at the same temperature. The effective decoherence rate of a PTI-initialized system is predicted to satisfy:

$$(9) \Gamma_{PTI} = \Gamma_{th} \cdot (1 - \eta_{adm})$$

where $\eta_{adm} \in (0, 1)$ is the admissibility efficiency of the PTI design, a dimensionless parameter determined by how closely the initialized state approaches the minimum-leakage phase configuration permitted by Axiom 4 [PT, Sec. XIX.3]. A PTI design of perfect admissibility efficiency would achieve $\Gamma_{PTI} = 0$, i.e., complete coherence preservation at any temperature — a prediction that Phase Theory does not make, because the phase-substrate axioms do not permit a perfectly admissibility-efficient PTI at finite temperature. But Phase Theory does predict measurably lower decoherence rates for high-efficiency PTI structures compared to generic initializations at the same physical temperature.

The experimental path for Test Class IV exploits the precision decoherence characterization capabilities of superconducting qubit platforms and trapped-ion systems. In both platforms, decoherence rates can be measured with high precision through Ramsey spectroscopy and dynamical decoupling benchmarking, and the physical temperature can be independently controlled and measured. The test protocol compares decoherence rates at fixed temperature for PTI-initialized states versus generic initializations versus standard dynamical decoupling sequences. The admissibility efficiency of each PTI design is computed from the Phase Theory formalism and used to generate a quantitative prediction for Γ_{PTI} . The experiment must be designed so that the PTI initialization does not alter the temperature or the system-bath coupling — only the phase-topological structure of the initial state — to isolate the admissibility-efficiency effect.

The falsification condition for Test Class IV is as follows: if decoherence rates are strictly bounded by temperature regardless of PTI design — that is, if no PTI structure achieves a measurably lower decoherence rate than the thermal baseline Γ_{th} at any temperature across all PTI designs tested — then Phase Theory is falsified on this prediction class. The absence of any admissibility-efficiency effect, even at the highest-efficiency PTI designs achievable with current quantum control technology, would eliminate the non-thermal phase memory prediction as an accessible empirical consequence of Phase Theory.

8. Test Class V: Non-Thermal Correlations in Black Hole Evaporation

The black hole information paradox, in its sharpest form [2, 3], asks whether the process of black hole formation and evaporation preserves unitarity. If Hawking radiation is exactly thermal — as the original semiclassical calculation suggests — then the information encoded in the quantum state of the infalling matter is irretrievably lost, violating the unitarity of quantum evolution. If, on the other hand, Hawking radiation carries subtle quantum correlations sufficient to encode the full quantum state of the original matter, then the radiation is not exactly thermal, and the mechanism by which information escapes must be identified. This paradox has resisted resolution for five decades, despite significant partial progress through the AdS/CFT correspondence [4] and more recent work on quantum error correction in holographic systems [24].

Phase Theory resolves the paradox by construction: information is not destroyed in black hole evaporation because it is holographically encoded in the phase topology of the horizon. In Phase Theory, the horizon is a coherence boundary — a surface at which the mutual information between the interior and exterior phase regions is maximally close to the saturation bound of Axiom 4. Hawking radiation, reinterpreted within Phase Theory as phase

leakage through this coherence boundary, is not exactly thermal: it carries non-thermal correlations that encode the phase-topological structure of the horizon and therefore preserve unitarity. The leading non-thermal correction to the Hawking radiation spectrum takes the form of a high-frequency excess relative to the Planck distribution, arising at the frequency scale ω_{coh} set by the coherence saturation boundary:

$$(10) \Delta n(\omega) = n_{\text{Hawking}}(\omega) \cdot \varepsilon_{\text{PT}}(\omega/\omega_{\text{coh}})$$

where ε_{PT} is a dimensionless function that vanishes for $\omega \ll \omega_{\text{coh}}$ and approaches a constant of order unity near $\omega \sim \omega_{\text{coh}}$ [PT, Sec. XXII.4]. This non-thermal correction is unobservably small for astrophysical black holes and becomes significant only near the Page time — the half-evaporation point at which the entanglement entropy of the radiation begins to decrease — and for very small black holes near the end of evaporation.

The primary experimental access point for Test Class V is through analogue black hole systems. Photonic crystal systems and Bose-Einstein condensates have been engineered to reproduce the Hawking effect at analog coherence scales where the non-thermal corrections are measurable [21]. In analogue black hole experiments, the analog Hawking temperature is set by the sound-horizon gradient in the condensate or the group-velocity horizon in the photonic crystal, and the analog coherence scale ω_{coh} is set by the healing length of the condensate or the photonic bandgap width. Phase Theory predicts that the radiation spectrum in these systems exhibits a non-thermal high-frequency excess parameterized by ε_{PT} — a prediction that is absent in the free-field Hawking calculation and in any local semiclassical description of the analog system.

A complementary astrophysical path, contingent on the detection of evaporating primordial black holes, involves correlation analysis of multi-messenger emission: Phase Theory predicts that the correlations between

photon energies in the late-stage evaporation spectrum exceed those predicted by any thermal emission model, with a characteristic energy scale set by the analog of ω_{coh} for the astrophysical system. The falsification condition for Test Class V is the following: if the evaporation spectrum in high-fidelity analogue black hole experiments is demonstrably perfectly thermal at all stages — with no high-frequency excess beyond the Planck distribution, within the sensitivity of the analog apparatus — then Phase Theory's holographic information encoding claim is directly contradicted, and the theory is falsified on this prediction class.

9. Test Class VI: Deviations from Λ CDM in Late-Time Expansion

The Λ CDM model of cosmology — the standard model of the universe, in which late-time expansion is driven by a cosmological constant Λ with equation of state $w = -1$ — provides an excellent fit to current observational data, from the CMB power spectrum measured by Planck [23] to the baryon acoustic oscillation scale measured by DESI [17]. The cosmological constant Λ is, however, a conceptual puzzle of the first order: its observed value is approximately 120 orders of magnitude smaller than the naive quantum field theory estimate of the vacuum energy density, a discrepancy that has no accepted resolution within the standard frameworks.

Phase Theory offers a structural resolution of this problem by reidentifying the cosmological constant not as a fundamental parameter of spacetime but as the background phase potential $V(\Phi_0)$ evaluated at the vacuum configuration of the phase substrate — the residual phase-inconsistency energy of the global vacuum [PT, Sec. XXIII.2]. This identification carries an immediate and testable consequence: unlike a true cosmological constant, the background phase potential is a dynamical quantity that evolves as the phase substrate continues to relax on cosmological timescales. The effective

equation of state of dark energy in Phase Theory is therefore not exactly $w = -1$ but a slowly evolving function:

$$(11) \ w(z) = -1 + \varepsilon_{\text{relax}}(z) \cdot f(z)$$

where $\varepsilon_{\text{relax}}(z)$ is a small, positive, redshift-dependent relaxation parameter encoding the rate of ongoing phase-substrate relaxation at redshift z , and $f(z)$ is a form factor derived from the Phase Theory dark energy dynamics [PT, Sec. XXIII.4]. Phase Theory predicts that $\varepsilon_{\text{relax}}(z)$ is correlated with the large-scale coherence metrics of cosmic structure — specifically, that the deviation of $w(z)$ from -1 is larger in redshift bins where the coherence of large-scale structure (as measured by galaxy two-point correlations) is lower, because phase relaxation proceeds more rapidly in less coherent regions of the phase substrate.

The recent DESI 2024 BAO results [17] have already reported hints of $w \neq -1$ at the $2\text{--}3\sigma$ level in certain redshift bins. Phase Theory predicts that this deviation, if real, should exhibit the specific redshift dependence and correlation with structure coherence metrics described by Equation (11) — a prediction that can be tested with the full DESI survey data, the Euclid wide-field survey, and ultimately the Nancy Grace Roman Space Telescope's deep field observations. The experimental protocol requires tomographic decomposition of the dark energy signal by redshift bin, cross-correlation with large-scale structure coherence metrics, and careful control for baryonic systematics and lensing contamination.

The falsification condition for Test Class VI is the following: if the expansion dynamics are fully consistent with a pure cosmological constant ($w = -1$) at all redshifts and scales accessible to next-generation surveys — and in particular if no correlation between dark energy equation-of-state deviations and large-scale coherence metrics is detected — then Phase Theory's dark

energy reinterpretation is disfavored by the data and should be regarded as empirically excluded.

10. Test Class VII: Phase-Derived Saturation of Information Capacity

The coherence limitation of Phase Theory (Axiom 4, Equation 4) imposes an area-scaling bound on the maximum information density achievable in a coherent quantum system. This is a holographic constraint [10, 11, 12]: information is encoded not in the volume of a coherent domain but in the area of its boundary, and the maximum information capacity of a region of phase substrate scales as its boundary area rather than its volume. In the standard quantum field theory description of a quantum register — a collection of N qubits in a volume V — the maximum information capacity scales as N , which scales with V at fixed qubit density. Phase Theory predicts that this volume scaling fails at sufficiently high information density, replaced by area scaling at and above the coherence saturation bound.

The experimental consequence of this prediction is accessible through quantum error correction. In a quantum error-correcting code, the logical error rate as a function of the physical error rate exhibits a threshold behavior: below a threshold physical error rate p_{th} , the logical error rate decreases exponentially with code distance, and above it, encoding makes the situation worse. The threshold p_{th} is a property of the code and the noise model. Phase Theory predicts that, as the physical qubit density ρ (qubits per unit volume) is increased at fixed physical error rate, the threshold $p_{th}(\rho)$ decreases — because the area-scaling information capacity bound of Axiom 4 limits the effective code distance available at high density. Specifically:

$$(12) \ p_{th}(\rho) = p_{th,0} \cdot [1 - (\rho / \rho_{sat})^{1/3}]$$

where $p_{th,0}$ is the zero-density threshold and ρ_{sat} is the coherence saturation density derived from the Phase Theory coherence bound [PT, Sec. XX.2]. This

formula predicts that the threshold collapses sharply as $\rho \rightarrow \rho_{\text{sat}}$ — a sharper-than-predicted threshold — and that ρ_{sat} scales with the boundary area of the coherent domain divided by its volume, i.e., as L^{-1} for a domain of linear dimension L . This area-versus-volume scaling of ρ_{sat} is the key experimental discriminant: it predicts that a larger coherent domain achieves a higher saturation density than a smaller one of the same volume, because the area-to-volume ratio is more favorable.

The experimental path for Test Class VII requires logical error rate measurements as a function of physical qubit density in superconducting and photonic quantum processors, with systematic variation of the coherent domain size. Current generation superconducting processors — in which physical qubit densities and code distances can be independently varied through chip layout — are approaching the regime where Phase Theory predicts this effect to become detectable, particularly as processors scale beyond 1,000 physical qubits per coherent domain. Comparison with QFT-based threshold predictions, which predict no ρ -dependence of p_{th} at any accessible density, provides a clean falsification criterion.

The falsification condition for Test Class VII is the following: if no information density saturation is observed — if error correction thresholds are independent of physical qubit density at all accessible densities, scale with volume rather than area as coherent domain size is varied, and no sharper-than-predicted threshold exists in any tested system — then Phase Theory's coherence limitation axiom is empirically contradicted and the theory is falsified on this prediction class.

11. Negative Predictions: What Phase Theory Asserts Will Not Be Observed

A theoretically accountable framework must not only specify what it predicts will be observed — it must also specify what it predicts will not be observed.

Negative predictions are, in some respects, more powerful falsifiers than positive ones: a single confirmed observation of a negatively predicted phenomenon constitutes an immediate and unambiguous refutation. Phase Theory makes four explicit negative predictions, each grounded in a structural feature of the theory's axioms, and each falsifiable by a specific class of experimental observation.

The first negative prediction concerns the particle spectrum. Phase Theory predicts approximately 25 ± 10 detectable particle species beyond the Standard Model at energies accessible to next-generation colliders — a bounded spectrum, not an open-ended proliferation [PT, Sec. X.2]. The predicted species are precisely those corresponding to stable and meta-stable topological defects in the target manifold $T = U(1) \times SU(2) \times SU(3)$: winding configurations, vortex structures, and kink solutions whose stability is guaranteed by the topology of the target space. Phase Theory does not admit particles in representations that are not stabilized by the topology of T — it does not, for example, predict magnetic monopoles in the standard sense (which would require a different target topology) or particles in exotic representations incompatible with the $SU(3)$ color symmetry. Discovery of a particle species at any next-generation collider whose properties are inconsistent with the Phase Theory topological classification scheme — in particular, a species in a group representation not admitted by T , or a species whose quantum numbers are incompatible with any stable or meta-stable defect configuration in T — would falsify Phase Theory immediately and decisively.

The second negative prediction concerns Lorentz invariance. Phase Theory asserts that Lorentz invariance is an emergent symmetry of the propagation of coherent phase modes in a flat background, broken only at energy scales approaching the phase-coherence scale E_{coh} — which is predicted to be near or above the Planck scale [PT, Sec. VII.3]. At all energies currently accessible

to experiment, including the highest-energy cosmic rays and the most energetic gamma-ray bursts, Lorentz violation in Phase Theory is predicted to be immeasurably small, suppressed by powers of (E/E_{coh}) . This prediction is in contrast with Loop Quantum Gravity, which predicts Lorentz violation effects potentially detectable in gamma-ray burst timing [8]. Any confirmed detection of Lorentz violation at energies below E_{coh} — at any level inconsistent with the suppression factor $(E/E_{\text{coh}})^n$ for any integer power n consistent with the Phase Theory dimensional emergence mechanism — would falsify Phase Theory's mechanism for the emergence of spacetime symmetry.

The third negative prediction is the absence of macroscopic superluminal signaling. This prediction follows from a structural theorem of Phase Theory — the No-Closed-Phase-Loop Theorem (Theorem 20.1 of [PT]) — which asserts that no self-consistent phase configuration can contain a closed causal loop. The maximum propagation speed of phase disturbances is the speed of light, which emerges as a phase-substrate parameter rather than a postulate [PT, Sec. VII.1]. No configuration of phase-substrate fields can transmit information faster than this maximum speed, regardless of the degree of entanglement or coherence engineering applied to the system. Any verified experimental instance of superluminal signaling — one that cannot be attributed to measurement error, loopholes in the experimental design, or misidentification of the signal — would falsify Phase Theory at the foundational level, because it would require a closed causal loop that the theory structurally forbids.

The fourth negative prediction is the impossibility of thermodynamic violation. Phase Theory derives the second law of thermodynamics from the monotonic directionality of phase relaxation: the phase-substrate functional $I[\Phi]$ is monotonically non-increasing under the relaxation dynamics of Equation (1), and this monotonicity is identified with the non-decrease of

entropy in isolated macroscopic systems [PT, Sec. XXI.2]. The second law in Phase Theory is therefore not a statistical approximation valid with high probability — it is a consequence of the irreversibility built into the phase relaxation equation at the fundamental level. Phase Theory predicts no regime — no temperature, energy, system size, or preparation procedure — in which the second law fails or in which a Maxwell's demon operating within the physical laws of the phase substrate could achieve a net decrease in entropy. Any experimental violation of the second law, at any scale, would falsify the monotonic phase relaxation assumption and with it the Phase Theory derivation of thermodynamics.

12. Risk Assessment: Where Phase Theory Is Most Vulnerable

An honest assessment of Phase Theory's empirical situation requires identifying not only its falsification conditions but also the probability that each falsification condition will be reached — that is, identifying the regimes in which the theory is most exposed. The following risk matrix organizes Phase Theory's principal empirical vulnerabilities by test class, vulnerability type, and current assessment of falsification risk.

Test Class	Prediction	Vulnerability	Risk Level
I — Randomness Collapse	$C_{\text{adm}}(\tau) \neq 0$ at saturation	Signal may be below current detection threshold; coherence engineering constraints	High
II — Measurement Memory	Non-Markovian admissibility correlations	Correlations may be indistinguishable from decoherence artifacts	High
III — Photon	Non-monotonic	Environmental	Medium

Test Class	Prediction	Vulnerability	Risk Level
Non-Paradox	visibility $V(L)$	decoherence may mask global phase signal at km scale	
IV — Phase Cooling	$\Gamma_{\text{PTI}} < \Gamma_{\text{th}}$	PTI design efficiency may be insufficient to produce measurable deviation	Medium
V — Black Hole Information	Non-thermal analogue radiation spectrum	Analog fidelity limitations; only indirect access to astrophysical case	High
VI — Cosmological Phase Drift	$w(z) \neq -1$ correlated with structure	Degeneracy with baryonic effects; observational systematic errors	Medium
VII — Information Density	Area-scaling error thresholds	Current processor densities may not yet reach ρ_{sat}	Lower (near-term scalable)

The risk matrix reveals that Phase Theory is most vulnerable in three regimes. First, in extreme coherence conditions required for Test Class I: the admissibility correlation function $C_{\text{adm}}(\tau)$ must be non-zero and detectable at achievable coherence depths for Phase Theory to survive as a predictive framework. If $C_{\text{adm}}(\tau)$ is non-zero in principle but undetectable at any coherence depth reachable with current or near-future technology, the theory's core claim — that randomness is apparent rather than irreducible — is empirically inaccessible, and the theory fails not by being falsified but by failing to make contact with experiment. This is a different kind of failure than falsification, but it is equally terminal for a scientific theory.

Second, Phase Theory is highly exposed in the long-duration correlated measurement experiments of Test Class II. If all sequential measurement correlations observed in weak-measurement chains are fully and quantitatively accounted for by standard decoherence models — with no residual signature after the decoherence contribution is subtracted — then the admissibility memory prediction is empirically invisible. The difficulty is that the decoherence contribution and the admissibility memory contribution are both small, and disentangling them requires the independent variation of decoherence rate and apparatus coupling strength that constitutes the key experimental control. If this independent variation proves technically impossible in accessible systems, Test Class II may be unresolvable at current technological levels.

Third, Phase Theory faces its most direct challenge in the analogue black hole experiments of Test Class V. This is the only test class that directly probes the holographic information encoding claim — the assertion that information is encoded in boundary phase topology rather than bulk volume — and a demonstrably thermal spectrum in a high-fidelity analogue experiment would be a severe result for the theory. Phase Theory is explicit that this risk is intentional: a theory that avoids placing its central holographic claim in the path of experiment is not a scientific theory of holography. The analogue black hole experiments currently in progress with Bose-Einstein condensates [21] are approaching the regime of sensitivity relevant for this test.

It is essential to note explicitly that this vulnerability profile is not a defect of Phase Theory — it is a feature. A theory with no experimental risks is a theory that makes no predictions, and a theory that makes no predictions is not a physical theory. Phase Theory stakes falsifiable claims in exactly the experimental territories that its structural commitments demand, and it accepts the corresponding risks as the cost of scientific accountability.

13. Falsifiability in Context: Comparison with Competing Frameworks

Phase Theory does not exist in isolation. Its claims about ontology, dynamics, and experimental prediction must be evaluated in the context of the existing landscape of foundational frameworks, each of which makes its own peace — or fails to — with the criterion of falsifiability. Four frameworks merit explicit comparison: the Everett (many-worlds) interpretation, string theory, dynamical collapse models, and loop quantum gravity.

The Everett interpretation of quantum mechanics [5] posits that the quantum state of the universe evolves unitarily under the Schrödinger equation at all times, and that what appears to be a measurement outcome is the branching of the universal wavefunction into non-interacting components, each of which contains an observer who perceives a definite result. The Everett interpretation is internally consistent and avoids the measurement problem by stipulation. However, it is, strictly speaking, non-falsifiable as a framework: it makes no prediction that differs from standard quantum mechanics in any accessible experiment, because it accepts the Born rule as an effective probability rule within each branch without explaining why that branch is experienced rather than another. Phase Theory, by contrast, makes specific deviations from Born-rule predictions in Test Classes I and II that are absent in any Everett-based framework. If those deviations are observed, they would confirm Phase Theory and falsify any framework — including Everett — that predicts exact Born-rule statistics. If they are not observed, Phase Theory is falsified while Everett remains consistent with the data.

String theory [19] represents arguably the most ambitious foundational program in contemporary physics, deriving gauge symmetries, matter content, and gravity from the dynamics of one-dimensional objects in a higher-dimensional spacetime. Its empirical situation, however, is complicated by the landscape: the approximately 10^{500} distinct string vacua

consistent with low-energy physics make string theory's experimental predictions ambiguous at currently accessible energies — almost any low-energy observation can be accommodated by an appropriate choice of vacuum [18]. Individual string models make specific predictions, but the theory as a whole lacks a mechanism for selecting the physical vacuum, and without such a mechanism, it cannot be falsified at the level of the theory rather than the model. Phase Theory has no landscape: the target manifold $T = U(1) \times SU(2) \times SU(3)$ is fixed by the axioms, and the particle spectrum is bounded (25 ± 10 species beyond the Standard Model), making Phase Theory falsifiable at next-generation colliders in a way that string theory as a whole is not.

Dynamical collapse models — particularly the GRW model [6] and its continuous stochastic generalization, the CSL model [7] — address the measurement problem by introducing modifications to the Schrödinger equation that cause wavefunction collapse to occur spontaneously at a rate controlled by two free parameters: the collapse rate λ and the localization length r_c . These parameters are chosen post hoc to make collapse effects sub-threshold in existing experiments while still producing definite outcomes on macroscopic timescales. Phase Theory and collapse models agree in predicting deviations from standard QM but disagree on the mechanism: collapse models predict spontaneous localization of the wavefunction, while Phase Theory predicts admissibility-memory correlations in sequential measurements with no spontaneous localization. Critically, these are distinguishable predictions in Test Class II: admissibility memory produces correlations across sequential measurements that depend on apparatus coupling strength and persist even when no spontaneous localization is predicted by collapse models. A positive Test Class II result would discriminate between Phase Theory and collapse models; a negative result would falsify Phase Theory while leaving collapse models viable (subject to their own experimental constraints).

Loop quantum gravity [8] approaches the quantization of spacetime by applying canonical quantization to general relativity, deriving a discrete spacetime structure at the Planck scale. One of LQG's most cited experimental predictions is Planck-scale Lorentz violation, detectable in principle through energy-dependent time delays in gamma-ray burst photons observed by the Fermi Gamma-ray Space Telescope and the upcoming Cherenkov Telescope Array. Phase Theory predicts no Lorentz violation at accessible scales, because Lorentz symmetry in Phase Theory is an exact symmetry of coherent phase mode propagation broken only at $E_{\text{coh}} \sim E_{\text{Planck}}$. These are qualitatively distinct predictions that could jointly constrain both theories through astrophysical observation: detection of an energy-dependent time delay consistent with LQG predictions would falsify Phase Theory's dimensional emergence mechanism, while the absence of such a delay would disfavor LQG's Lorentz violation prediction while remaining consistent with Phase Theory.

14. The Experimental Moment: Why These Tests Are Accessible

It is a remarkable and non-trivial fact that Phase Theory arrives in a moment when several of its key experimental tests are accessible, or near accessible, with current technology. This is not a contrivance — Phase Theory did not choose its predictions to match available experimental capabilities. The predictions follow from the axioms, and the axioms were formulated without reference to current experimental program. That the predictions happen to land in accessible territory is, if anything, a constraint on the theory: a theory that proposes tests accessible only in principle, at energy scales or coherence depths unachievable for the foreseeable future, is less scientifically accountable than one whose tests are within experimental reach. Phase Theory accepts the constraint of accessibility as a positive feature.

In the domain of high-coherence photonics, modern platforms — including integrated photonic circuits, fiber-based interferometers, and free-space optical systems with active phase stabilization — achieve coherence times sufficient to begin exploring the admissibility correlation function $C_{\text{adm}}(\tau)$. Photon-number-resolving detectors with efficiency exceeding 90% are now commercially available, and phase control at the sub-shot-noise level has been demonstrated in multiple laboratory settings. The primary technical challenge for Test Classes I and III is not detector quality but sequence length and stability: generating and analyzing outcome sequences of $N > 10^8$ events per coherence-depth setting over extended periods requires sustained experimental control that is at the frontier of current capability but not beyond it.

In quantum error correction, superconducting quantum processors from multiple groups now achieve logical qubit densities sufficient to begin probing the information density saturation predicted in Test Class VII. The recent demonstration of below-threshold logical error rates in superconducting surface codes, with physical qubit counts exceeding 1,000 and code distances sufficient for meaningful threshold measurements, provides a direct experimental platform for the area-versus-volume scaling test. As processor scales increase toward 10^4 – 10^5 physical qubits per coherent domain — a near-term roadmap target for multiple leading groups — the Phase Theory prediction of a density-dependent threshold shift becomes increasingly discriminating.

In precision cosmology, the DESI Bright Galaxy Survey and the full DESI Year-5 dataset will provide measurements of the baryon acoustic oscillation scale and the dark energy equation of state with sufficient precision to resolve sub-percent deviations from $w = -1$ in multiple redshift bins [17]. The Euclid wide-field survey, launched in 2023, will add weak lensing tomography and galaxy clustering measurements across the full redshift range $0 < z < 2$,

providing the large-scale structure coherence metrics needed to test the correlation predicted in Equation (11). The Nancy Grace Roman Space Telescope's deep field observations will extend this reach to higher redshifts. Together, these surveys provide the statistical power to make Test Class VI decisive within the next decade.

In analog gravity, the experimental program on analogue black holes using Bose-Einstein condensates has advanced substantially [21]. Experiments in multiple groups have confirmed the analog Hawking effect at temperatures consistent with the surface gravity of the sonic horizon, and the precision of radiation spectrum measurements is improving with each generation of apparatus. The Phase Theory prediction of a non-thermal high-frequency excess in the radiation spectrum lies within the sensitivity range of next-generation analogue systems operating with tighter control over condensate density and horizon sharpness. Phase Theory did not exist when most of these platforms were designed. That it arrives when it can be tested — rather than a century hence — is a fact about the theory's predictions, not about its politics. A theory that proposes these tests must survive them.

15. Definitive Failure Conditions for Phase Theory

Phase Theory is falsified — not merely disfavored, not merely in tension with data, but definitively falsified — if any of the following five conditions is confirmed by experiment. These conditions are specified without qualification or room for reinterpretation. They are stated here as commitments of the theory, not as provisional claims subject to adjustment in the light of contrary evidence.

The first definitive failure condition is that admissibility leaves no empirical trace: specifically, that $C_{\text{adm}}(\tau) = 0$ at all accessible coherence depths across all Test Class I experiments, with sufficient statistical power to rule out the Phase Theory prediction at the 5σ level. If randomness is found to be

irreducible at all scales and under all conditions of coherence engineering — if Born-rule statistics hold exactly with no admissibility-correlated deviations across all Test Class I and II experiments — then Phase Theory's foundational claim that the apparent randomness of quantum mechanics is an emergent property of a deterministic phase substrate is empirically inaccessible, and the theory is not science.

The second definitive failure condition is that sequential measurement correlations beyond standard QM are entirely absent: that all correlations across sequential measurements in all accessible weak-measurement chain experiments are fully and quantitatively accounted for by standard quantum mechanics augmented by standard decoherence theory, with no residual non-Markovian admissibility signature surviving systematic variation of apparatus coupling strength. This would eliminate the admissibility memory prediction — and with it the Phase Theory account of the measurement process — as having any empirical content whatsoever.

The third definitive failure condition is that collapse is experimentally required: that a dynamical collapse process, independent of decoherence and inconsistent with unitary evolution, is experimentally confirmed as necessary to account for measurement outcomes. This would directly contradict Phase Theory's replacement of collapse by emergent decoherence, and would validate the collapse model framework [6, 7] at the expense of Phase Theory. The threshold for this condition is the experimental confirmation of spontaneous localization at a rate and localization scale inconsistent with any decoherence model but consistent with the GRW or CSL parameter regime.

The fourth definitive failure condition is that coherence engineering produces no non-thermal effects: if phase-topological initialization structures in Test Class IV produce decoherence rates indistinguishable from thermal baselines at all temperatures and all PTI designs, across all tested platforms, then the admissibility efficiency prediction of Equation (9) is empirically null and the

non-thermal phase memory mechanism has no observational consequence. This failure condition is perhaps the most technically accessible of the five, because it requires only precision decoherence characterization at fixed temperature with and without PTI structures — an experiment well within the current capability of superconducting qubit and trapped-ion platforms.

The fifth definitive failure condition is that information density saturation is volume-scaling: if error correction thresholds in Test Class VII are found to be independent of physical qubit density at all accessible densities, scale with volume rather than area as coherent domain size is varied, and exhibit no sharper-than-predicted threshold at any density in any tested system, then the coherence limitation axiom (Axiom 4, Equation 4) is empirically contradicted. Since the coherence limitation is the axiomatic source of holography, the area-law entropy of black holes [10], and the Phase Theory resolution of the information paradox, a volume-scaling failure of Test Class VII would be devastating for the entire theoretical structure, not merely for one prediction class.

These five failure conditions are clear, unambiguous, and non-negotiable. They cannot be accommodated by parameter adjustment, because Phase Theory has no adjustable parameters in the relevant sectors: the admissibility correlation function, the admissibility memory length, the admissibility efficiency, and the coherence saturation density are all determined by the phase-substrate axioms. They cannot be reinterpreted away, because each specifies an exact prediction — $C_{\text{adm}}(\tau) = 0$, $\Gamma_{\text{PTI}} = \Gamma_{\text{th}}$, p_{th} volume-scaling — whose experimental violation would be beyond dispute. Phase Theory's authors enumerate these conditions explicitly as commitments of the theory and as the terms on which it accepts the judgment of experiment.

16. Conclusion

Phase Theory does not protect itself from experiment. This is its most important feature as a scientific framework, and it is worth stating plainly. The theory proposes a radical ontological revision — the replacement of fields, spacetime, and irreducible randomness by a single phase-bearing substrate governed by coherence and admissibility constraints — and it accepts, as the price of that ambition, the obligation to specify precisely how it can be shown to be wrong. The seven test classes enumerated in this paper are not post hoc rationalizations of known results. They are forward-looking commitments generated by the axiomatic structure of Phase Theory and refined by the requirement that they be experimentally accessible with current or near-future technology.

The structure of the paper's argument can be summarized as follows. Phase Theory predicts empirical equivalence with established frameworks in all their established domains of validity — this is a hard constraint imposed by the axioms, not a convenience. It predicts genuine deviations only at the admissibility frontier — in four specific experimental regimes characterized by coherence saturation, distinguishability limits, defect-core structure, and non-thermal phase memory. The seven test classes operationalize these deviations into specific, falsifiable experimental predictions with explicit falsification conditions. The four negative predictions specify phenomena that Phase Theory structurally excludes and whose observation would constitute immediate refutation. The risk matrix acknowledges honestly that the theory is most exposed in Test Classes I, II, and V — precisely the classes that test its most foundational claims. The comparison with competing frameworks demonstrates that Phase Theory is more falsifiable, not less, than the alternatives currently dominating the foundational physics landscape.

The experimental moment for Phase Theory has arrived. High-coherence photonic platforms, precision quantum error correction processors, next-

generation cosmological surveys, and high-fidelity analogue gravity systems are all now operating in or approaching the regimes in which Phase Theory's predictions depart from those of standard frameworks. Test Classes I and II — admissibility structure in outcome statistics and sequential measurement memory — are the most immediately accessible, requiring only sustained photonic coherence and precision weak-measurement chains, both within current experimental capability. Test Class IV — admissibility-engineered decoherence reduction — requires only precision decoherence benchmarking at fixed temperature in current-generation superconducting or trapped-ion systems.

Phase Theory calls for experimental programs targeting all seven test classes, with priority given to Test Classes I, II, and IV as the nearest-term and most discriminating. If Phase Theory survives these tests — if admissibility structure is found in outcome statistics, if admissibility memory is found in sequential measurements, if PTI structures reduce decoherence beyond the thermal limit, and if information density thresholds exhibit area rather than volume scaling — it will have earned, through experiment rather than argument, its place in the canon of fundamental physics. If it does not survive them, it should be discarded. That is the standard Phase Theory accepts.

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